

Question	Answer	Marks
7(a)	(induced) e.m.f. is (directly) proportional to rate	M1
	of change of (magnetic) flux (linkage)	A1
7(b)(i)	flux = e.m.f. $\times$ time	C1
	flux = 0.54×15 = 8.1 Wb	A1
7(b)(ii)	$\Phi = BA$	C1
	area = $8.1 / (38 \times 10^{-6})$ = $2.1 \times 10^5 \text{ m}^2$	A1
7(b)(iii)	area = speed $\times$ time $\times$ width	C1
	$v = (2.1 \times 10^5) / (15 \times 68)$ = 210 m s^{-1}	A1
7(b)(iv)	opposing force (due to current in wings) must be backwards	B1
	from Fleming's left-hand rule, current (in wings) must be from Q to P	B1
	current is from – to + inside an e.m.f. source so P is at higher potential	B1

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